well as HIV may be a major predisposing factor for perinatal transmission of HCV.[2,9] However, in our setting, none of the 16 drug-using women without HIV transmitted infection to their babies. HCV-RNA levels of drug-using mothers with HIV coinfection were significantly higher than in those with HCV alone, although the ranges overlapped. Maternal viraemia and rate of HCV infection in the infant has been reported to be correlated.[10] Active maternal liver disease may also enhance the risk of viral HCV transmission.[2,10] In our study, 4 of the 8 mothers of infected babies had been diagnosed with chronic active hepatitis, but 4 had normal alanine aminotransferase at delivery. However, since histological diagnoses were not available for the 4 transmitting mothers with normal enzyme at delivery, they may have been highly infectious healthy carriers or had chronic liver disease in remission phase.

Although our study was small, we did not find any association between any specific maternal HCV genotype and an increased rate of neonatal infection.

The finding that vertical/perinatal transmission is enhanced when the mother is co-infected with HCV and HIV has important public health implications, especially in urban areas in which most drug abusers are infected with both viruses.

Participants in the study group: S Bresciani, M G Marin, D Padula, A Rodella (Brescia); I Bulgarelli, F Chiodo, E Magliano, G Miotto, M L Muggiasca, E Pilloton, R Pozzoli, F Pregliasco, L Romanò, C Stringhi (Milan); F D'Agostino, F Paolillo (Lodi); and B Zapparoli (Monza). This work was supported in part by a grant from the Italian Ministry of Health, Centro Studi, Rome. We thank Dr L Stuyver for supplying part of the reagents for genotyping.

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Institute of Virology (Prof A R Zanetti PhD, E Tanzi PhD) **and Paediatrics Department IV, University of Milan** (S Paccagnini MD), Prof N Principi MD); **Institute of Chemistry, Medical School, University of Brescia** (G Pizzocolo BSc); **Department of Obstetrics and Gynaecology, Hospital "L Mangiagalli", University of Milan** (M L Caccamo MD); **Department of Paediatrics, Hospital "Niguarda-Cà Granda", Milan** (Prof E D'Amico MD); **Blood Transfusion Centre, Hospital "Maggiore", Lodi** (G Cambiè MD); **and Laboratory of Clinical Pathology, Hospital "S Gerardo", Monza, Italy** (L Vecchi MD)

Correspondence to: Prof A R Zanetti, Institute of Virology, University of Milan, Via C Pascal 38, 20133 Milan, Italy

Effect of neonatal circumcision on pain responses during vaccination in boys

Anna Taddio, Morton Goldbach, Moshe Ipp, Bonnie Stevens, Gideon Koren

Using data from one of our randomised trials, we investigated post-hoc whether male neonatal circumcision is associated with a greater pain response to routine vaccination at 4 or 6 months. Pain response during routine vaccination with diphtheria-pertussis-tetanus (DPT) alone or DPT followed by *Haemophilus influenzae* type b conjugate (HIB) was scored blind. 42 boys received DPT and 18 also received HIB. After DPT, median visual analogue scores by an observer were higher in the circumcised group (40 *vs* 26 mm, p=0·03) After HIB, circumcised infants had higher behavioural pain scores (8 *vs* 6, p=0·01) and cried longer (53 *vs* 19 s, p=0·02). Thus neonatal circumcision may affect pain response several months after the event.

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In our randomised controlled trial of a topical anaesthetic cream (EMLA, eutectic mixture of local anaesthetics; Astra) on infant pain in routine diphtheria-pertussis-tetanus (DPT) vaccination, boys had higher pain scores than girls.[1] If this sex difference is a real effect, it may be partly related to previous experience with acute pain, such as circumcision. Circumcised babies have short-term alterations in behaviour, sleep patterns, frequency of feeding, crying, fussiness, and heart rate.[2–7] Effects beyond the first few hours after the painful event have not been investigated. We have done a post-hoc analysis from our trial[1] to investigate whether circumcision was associated with pain scores during vaccination.

Healthy boys aged 4–6 months (mean 5) who were vaccinated with DPT alone, or DPT and *Haemophilus influenzae* type b conjugate (HIB), were included. DPT was administered intramuscularly on the upper thigh 60 min after treatment with EMLA or placebo. Infants who also received HIB were given an intramuscular dose several minutes after the DPT and on the opposite leg without EMLA or placebo. All injections were made by one of two paediatricians.

During the DPT injection, an observer and the paediatrician who administered the vaccine (both unaware of treatment allocations) rated the infant's pain response on a 100 mm ungraded visual analogue scale (VAS: 0=no pain, 100=worst possible pain). We videotaped the infants during their one or two injections, until they settled. A coder who was unaware of the treatments rated the pain responses for all vaccinations on a modified behavioural pain scale (face, cry, and body movements). The total pain score was obtained by adding the scores of the components and varied between 0 and 10.[1] Baseline pain scores, post-procedural scores, net pain (ie, difference between baseline and post-vaccination score), and duration of cry were assessed for all injections. Infants who had undergone painful procedures that were not routine (eg, lumbar puncture, surgery) were excluded from analyses. Infants who were in discomfort before the vaccine (ie, baseline behavioural pain scores greater than 2) were also excluded from analysis of pain for that vaccination. Statistical differences were calculated with Mann-Whitney, χ^2, or Fisher's exact tests. Correlations were done with Spearman's method.

Of the 42 boys who received DPT, 30 (71%) had been circumcised. No significant differences were found between circumcised and uncircumcised infants in

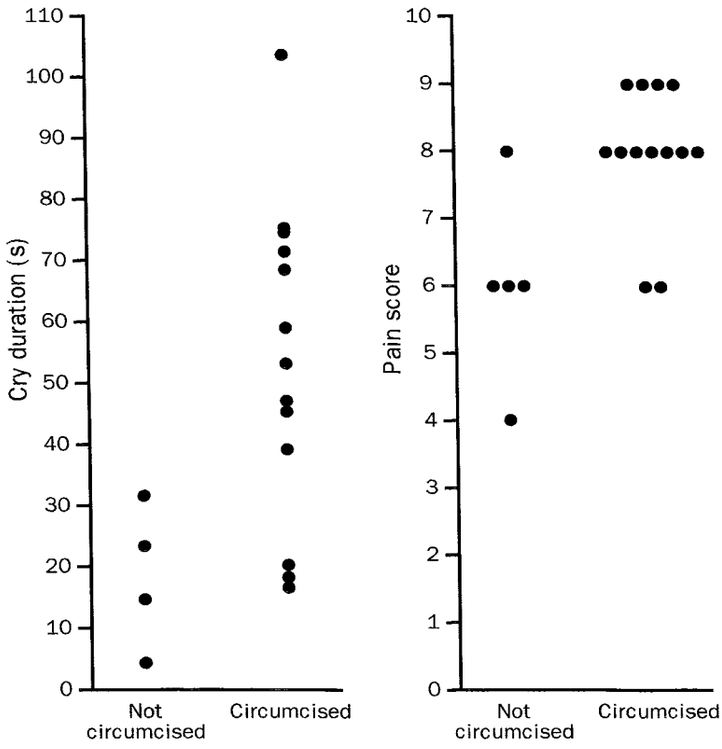


Figure: **Pain response during HIB injection in circumcised and uncircumcised boys**

Data not available for cry on 1 infant. p=0·02 for cry, p=0·01 for behavioural pain score.

demographic characteristics. Observer's VAS pain ratings were higher for the circumcised boys (median 40 *vs* 26 mm, p=0·03). Paediatricians' VAS scores showed a similar trend but were not significant (56 *vs* 30 mm, p=0·07). There was also a trend towards less crying for the uncircumcised boys (7·3 *vs* 22·3 s, p=0·06). The median post-vaccination behavioural pain score was 8 for the circumcised group compared with 7 in the uncircumcised group (p=0·2).

We stratified the infants by whether they were premedicated with the local anaesthetic (n=24) or placebo (n=18), and repeated the analyses. In the local anaesthetic group, both observer's and paediatricians' VAS scores were higher for circumcised boys (32 *vs* 10 mm, p=0·004; and 35 *vs* 13 mm, p=0·01, respectively). The total duration of crying was longer (14 *vs* 5 s, p=0·004), and there was a trend toward higher post-vaccination pain scores (7 *vs* 6, p=0·1) in the circumcised group. These results suggest that the local anaesthetic was more effective on infants that were uncircumcised (ie, may not already be conditioned to pain). No significant differences were observed in the pain responses of the placebo-treated infants.

18 infants received the HIB vaccine; 13 (72%) had been circumcised. These infants were younger than the other boys who participated in the trial (126 *vs* 140·5 days, p=0·02). They did not differ, however, in other demographic characteristics, or in pain scores and cry duration for the first vaccine. Circumcision was positively associated with post-vaccine behavioural pain scores (r=0·61, p=0·007), net pain scores (r=0·67, p=0·002), and duration of crying (r=0·57, p=0·02). Circumcised boys had higher scores and cried for longer (figure).

We also looked at cultural background (eg, being Jewish) and maternal intervention on pain response. No significant associations were found. Since 11 infants were initially excluded from the HIB analyses because of discomfort before injection, which may have introduced bias, analyses were repeated including these infants: the effects of circumcision status on pain response were even more statistically significant.

Male circumcision is the most common neonatal surgical procedure. It causes intense pain and measurable changes in behaviour that last up to 1 day. We found that circumcision status was associated with increased infant pain response to routine vaccination at 4–6 months. Circumcised boys had significantly longer crying bouts and higher pain scores. That both outcome measures, pain index, and cry duration, were influenced by circumcision status lends credibility to our observations. During the second (HIB) vaccination, circumcision status was more clearly associated with the observed pain response than after DPT. The DPT injection might have had a priming effect in circumcised infants which led them to exhibit even more pain after the HIB injection. The effects of memory and reinforcement on later nociceptive experience in neonates are not known.[8] Because memory of pain is believed to be important in subsequent pain perception, and the main structures for memory are functional in the neonatal period, it is conceivable that pain from circumcision may have long-lasting effects[9] on pain response and/or perception.

The post-hoc nature of our analyses and the small sample sizes make our conclusions speculative. Nevertheless, we suggest that analgesia should be routine for circumcision to avoid possible long-term effects in infant boys' pain responses.

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Department of Pediatrics, Hospital for Sick Children, University of Toronto, Toronto, Canada (A Taddio MSc, M Goldbach MD, M Ipp MD, B Stevens PhD, G Koren MD)

Correspondence to: Dr Gideon Koren, Division of Clinical Pharmacology and Toxicology, Hospital for Sick Children, 555 University Avenue, Toronto, Ontario, Canada M5G 1XG